

CSL-351: Computer Networks



Lab - 12

Cisco Packet Tracer (DHCP, DNS & Web Server)

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Content

- ❑ Dynamic Host Configuration Protocol (DHCP)
- ❑ Configuration of DHCP Protocol in network with packet tracer
- ❑ Domain Name Resolution (DNS)
- ❑ Configuration of DNS with packet tracer
- ❑ Web Server
- ❑ Configuration of Web Server with packet tracer

DHCP and DNS

- Dynamic Host Configuration Protocol (DHCP) is a client/server protocol that automatically provides an Internet Protocol (IP) host with its IP address and other related configuration information such as the subnet mask and default gateway
- The DHCP server maintains a pool of IP addresses and leases an address to any DHCP-enabled client when it starts up on the network. Because the IP addresses are dynamic (leased) rather than static (permanently assigned), addresses no longer in use are automatically returned to the pool for reallocation.

DHCP and DNS



IP Address =

Dynamic Host Configuration Protocol.

Every computer on a network has to have an I.P. address.

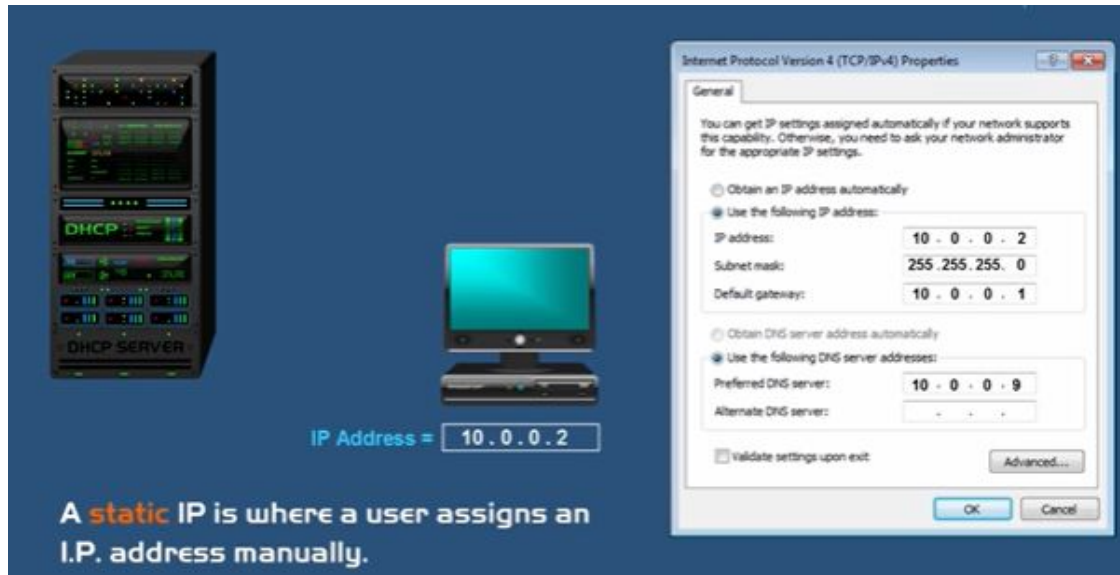
2 ways that a computer can be assigned an I.P. address.



Static IP or Dynamic IP.

DHCP and DNS

- Now a static IP is where a user assigns a computer or device with an IP address manually. Now this was the original method that was done in the beginning of networking. So as an example here we have a network connection properties window open for the network interface card on a Microsoft Windows computer

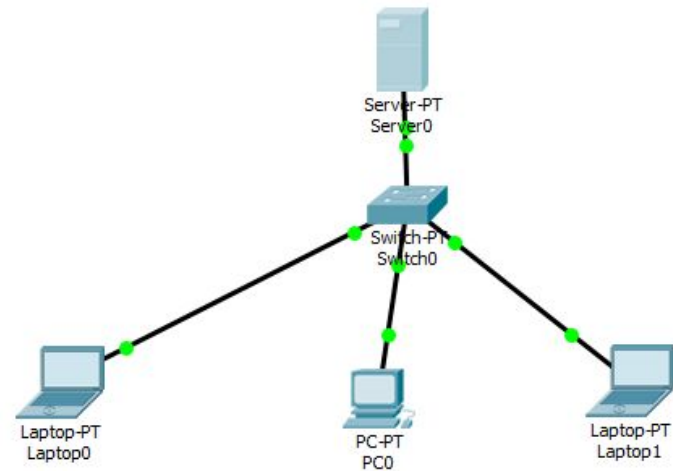


Why DHCP ?

- If you are dealing with a large network that has a lot of computers. And you also have to make sure that all the IP addresses are unique because if you assign the same IP address twice it would cause an IP conflict and would cause those computers to not have access to the network
- This computer is set to obtain an IP address automatically. So when you choose this option the computer would broadcast a request for an IP address on the network then the DHCP server will assign an IP address from its pool and deliver it to the computer.
- And then once that's done you can verify all the different settings that the DHCP server has given your computer. And you can do this by opening up a command prompt on a Windows computer and then type in `ipconfig /all` and then press "enter"

Configuration of DHCP protocol in a Network

- Create single network
- Required components
 - 3 end devices (1server, 3pc)
 - 1 switch



Give ip in the server

Server1

Physical Config Services Desktop Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IP Address 192.168.0.1

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

Link Local Address FE80::201:C7FF:FE53:4C19

IPv6 Gateway

IPv6 DNS Server

In the server > Go to Service > DHCP

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 192.168.0.1

DNS Server: 10.0.0.1

Start IP Address : 192 168 0 0

Subnet Mask: 255 255 255 0

Maximum Number of Users : 255

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.0.1	10.0.0.1	192.168.0.0	255.255.255.0	255	0.0.0.0	0.0.0.0

Go to PC > Go to IP > Select DHCP

PC1

Physical Config Desktop Programming Attributes

IP Configuration

☒ DHCP ☐ Static

IP Address 192.168.0.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.0.1

DNS Server 10.0.0.1

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

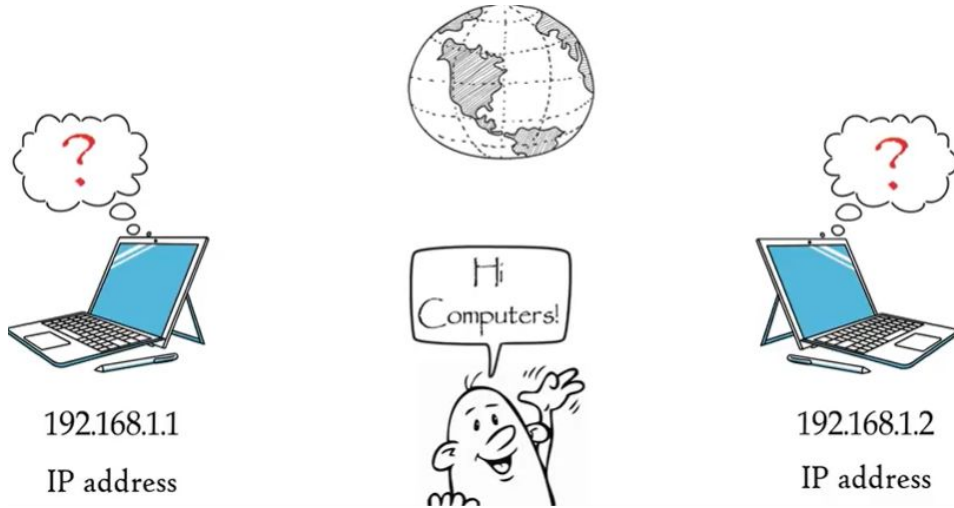
IPv6 Address /

Link Local Address FE80::201:42FF:FEBA:A909

IPv6 Gateway

IPv6 DNS Server

DNS (Domain Name System)



Computers always identify each other with unique numbers called IP addresses as they do not understand human languages

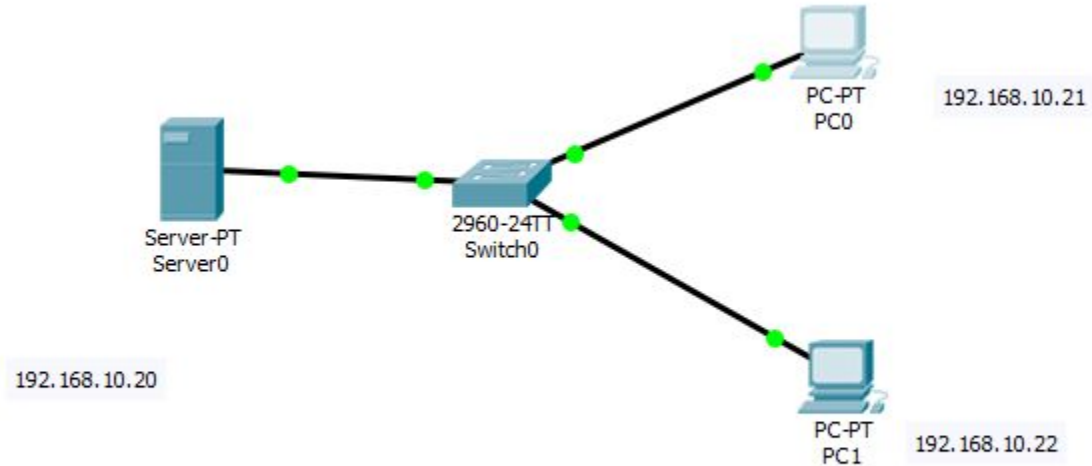
DNS (Domain Name System)

- DNS maintains a table where names are mapped into numbers or more specifically where websites domain names are mapped to their IP addresses
- Now when we type google.com in the web browser DNS translates it to the web browsers language that is IP address and gives it to the web browser
- Now web browser understands that we want to access Google.com so it contacts the Google server and loads google.com on the computer ensure DNS acts as a phone book of the Internet via computer

Configuration of DNS using packet tracer

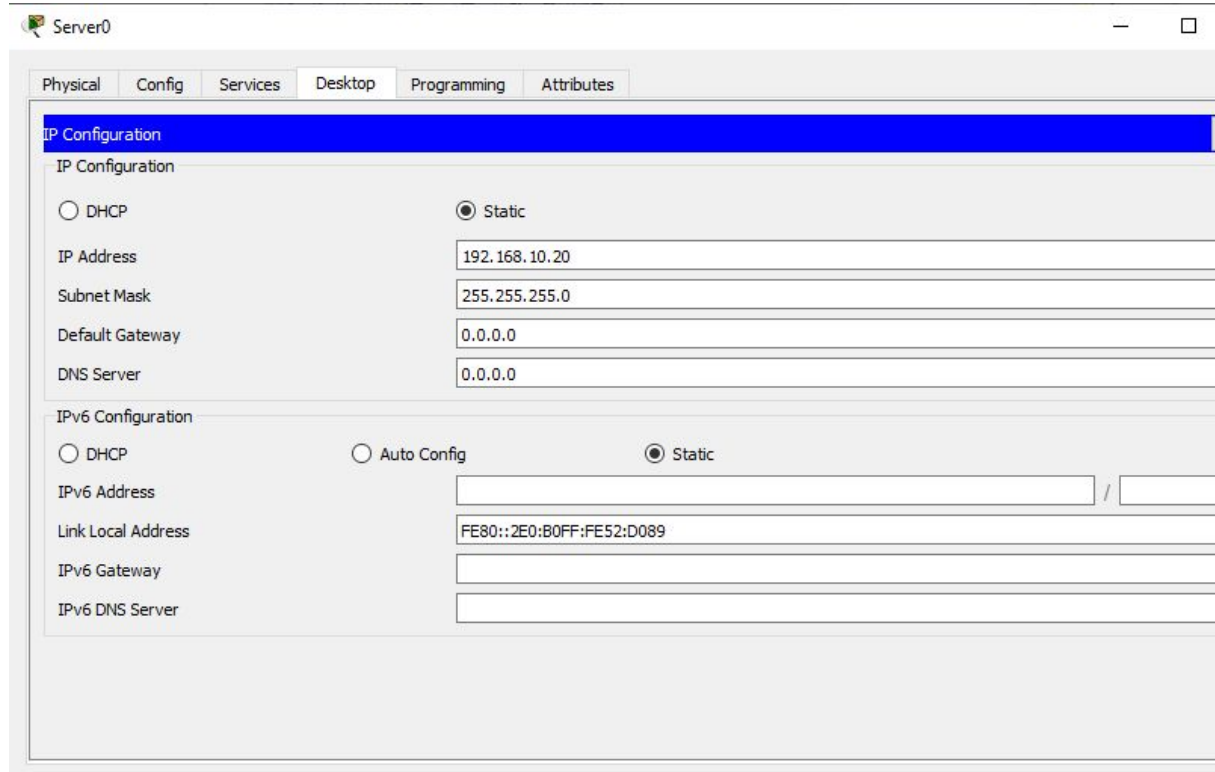
- Create single network
- Required components
 - 2 end devices , 1 Server
 - 1 switch (2960)

Place the devices



Steps

Assign IP to server and level it



The screenshot shows the 'IP Configuration' window for 'Server0'. The window has tabs for 'Physical', 'Config', 'Services', 'Desktop', 'Programming', and 'Attributes'. The 'Config' tab is active, and the 'IP Configuration' section is highlighted in blue. Below this, there are two main sections: 'IP Configuration' and 'IPv6 Configuration'.

IP Configuration

- ☐ DHCP
- ☒ Static
- IP Address: 192.168.10.20
- Subnet Mask: 255.255.255.0
- Default Gateway: 0.0.0.0
- DNS Server: 0.0.0.0

IPv6 Configuration

- ☐ DHCP
- ☐ Auto Config
- ☒ Static
- IPv6 Address: [Empty field] / [Empty field]
- Link Local Address: FE80::2E0:B0FF:FE52:D089
- IPv6 Gateway: [Empty field]
- IPv6 DNS Server: [Empty field]

Steps

Assign IP to all PC's and level it

PC0

Physical Config Desktop Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IP Address 192.168.10.21

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 192.168.10.20

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

Link Local Address FE80::230:A3FF:FEDE:7365

IPv6 Gateway

IPv6 DNS Server

Steps

Assign IP to all PC's and level it

PC1

Physical Config Desktop Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IP Address 192.168.10.22

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 92.168.10.20

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:58FF:FEB9:D96B

IPv6 Gateway

IPv6 DNS Server

Steps

The screenshot shows the 'Server0' configuration window with the 'Services' tab selected. The 'DNS' service is turned 'On'. Under 'Resource Records', there are three entries: 'cisco.com' (A Record, 192.168.10.20), 'pc0' (A Record, 192.168.10.21), and 'pc1' (A Record, 192.168.10.22). The 'DNS Cache' button is visible at the bottom.

SERVER0

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS**
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type **A Record**

Address

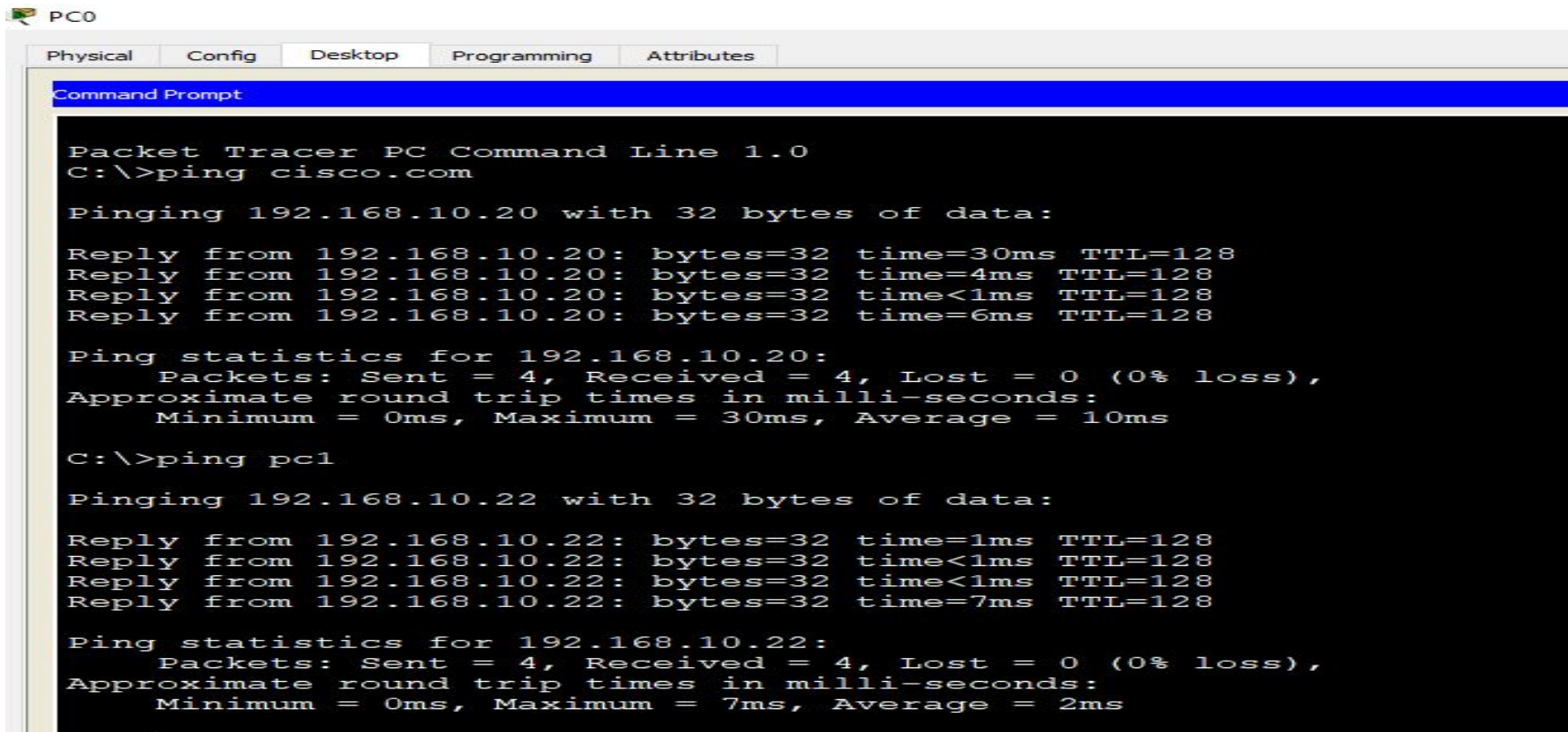
Add **Save** **Remove**

No.	Name	Type	Detail
0	cisco.com	A Record	192.168.10.20
1	pc0	A Record	192.168.10.21
2	pc1	A Record	192.168.10.22

DNS Cache

☐ Top

Output



The screenshot shows a Packet Tracer PC Command Line window for PC0. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Command Prompt shows the execution of two ping commands. The first command is 'ping cisco.com', which results in four successful replies from 192.168.10.20 with varying times (30ms, 4ms, <1ms, 6ms) and a TTL of 128. The ping statistics show 4 packets sent, 4 received, 0% loss, and an average round trip time of 10ms. The second command is 'ping pc1', which results in four successful replies from 192.168.10.22 with times (1ms, <1ms, <1ms, 7ms) and a TTL of 128. The ping statistics show 4 packets sent, 4 received, 0% loss, and an average round trip time of 2ms.

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt

Packet Tracer PC Command Line 1.0
C:\>ping cisco.com

Pinging 192.168.10.20 with 32 bytes of data:

Reply from 192.168.10.20: bytes=32 time=30ms TTL=128
Reply from 192.168.10.20: bytes=32 time=4ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time=6ms TTL=128

Ping statistics for 192.168.10.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 30ms, Average = 10ms

C:\>ping pc1

Pinging 192.168.10.22 with 32 bytes of data:

Reply from 192.168.10.22: bytes=32 time=1ms TTL=128
Reply from 192.168.10.22: bytes=32 time<1ms TTL=128
Reply from 192.168.10.22: bytes=32 time<1ms TTL=128
Reply from 192.168.10.22: bytes=32 time=7ms TTL=128

Ping statistics for 192.168.10.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 2ms
```

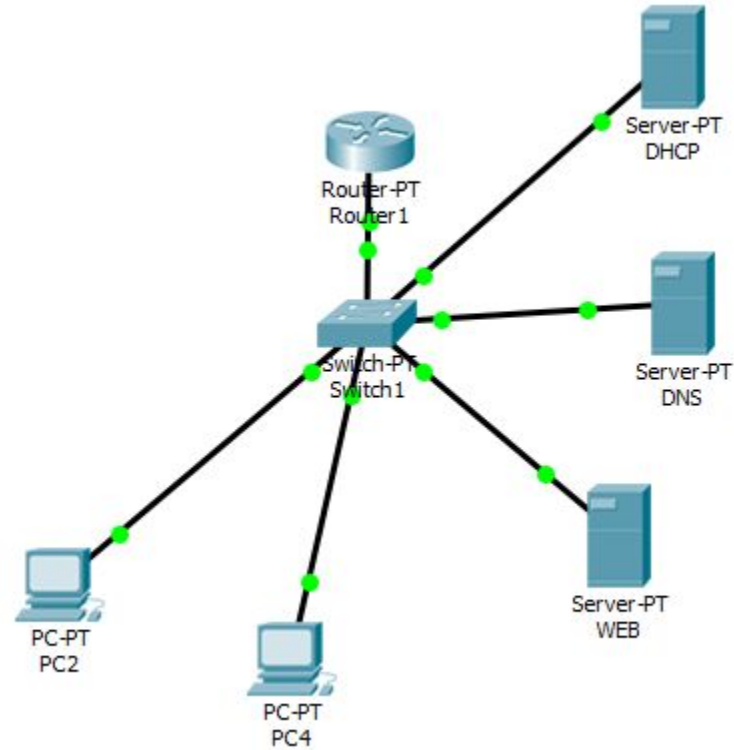
Configure Web Server

- Web Server: Web server is a program which processes the network requests of the users and serves them with files that create web pages. This exchange takes place using Hypertext Transfer Protocol (HTTP).
- Basically, web servers are computers used to store HTTP files which makes a website and when a client requests a certain website, it delivers the requested website to the client.
- For example, you want to open Facebook on your laptop and enter the URL in the search bar of google.
- Now, the laptop will send an HTTP request to view the facebook webpage to another computer known as the webserver. This computer (web server) contains all the files (usually in HTTP format) which make up the website like text, images, gif files, etc.
- After processing the request, the web server will send the requested website-related files to your computer and then you can reach the website.

Configuration of Web server using packet tracer

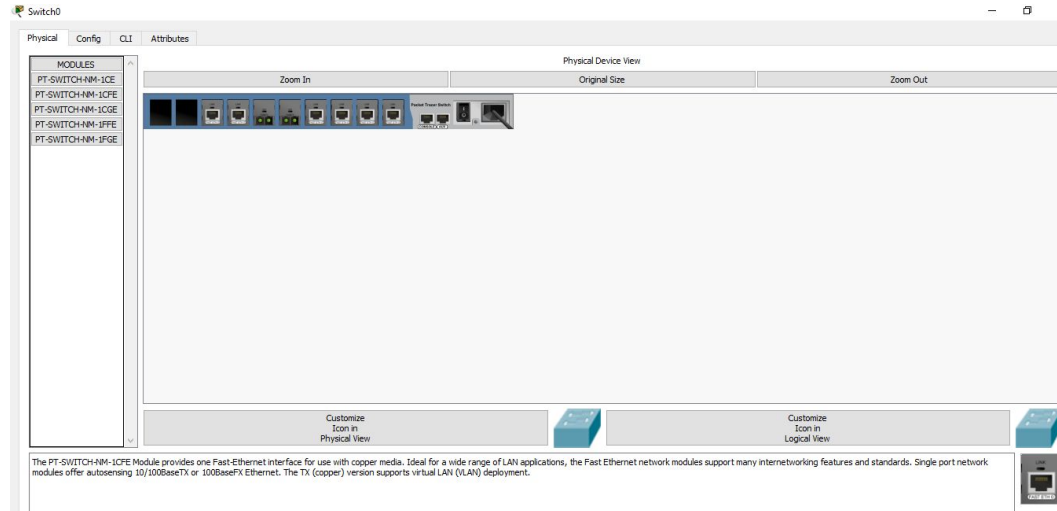
- Create single network
- Required components
 - 2 end devices , 3 Server
 - 1 switch (switch PT)
 - 1 router (Router PT)

Place the devices



Add port to the switch

- First switch off the SWITCH
- Add ports
- Switch on the devices



Configure Router

Router0

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet0/0

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☒ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 0060.7032.EC71

IP Configuration

IP Address 192.168.1.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Equivalent IOS Commands

```
changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet0/0, changed state to
up
```

☐ Top

Add IP to DHCP server

Server0

Physical Config Services Desktop Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IP Address 192.168.1.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 192.168.1.3

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

Link Local Address FE80::250:FFF:FE84:D099

IPv6 Gateway

IPv6 DNS Server

☐ Top

Add IP to DNS server

Server1

Physical Config Services Desktop Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IP Address 192.168.1.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 192.168.1.3

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

Link Local Address FE80::202:4AFF:FEC5:8C08

IPv6 Gateway

IPv6 DNS Server

☐ Top

Assign IP's to devices

PC0

Physical Config Desktop Programming Attributes

IP Configuration

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IP Address 192.168.1.4

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 192.168.1.3

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

Link Local Address FE80::230:A3FF:FE29:631D

IPv6 Gateway

IPv6 DNS Server

Top

PC1

Physical Config Desktop Programming Attributes

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IP Address: 192.168.1.5

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 192.168.1.3

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address: /

Link Local Address: FE80::2E0:B0FF:FE53:7917

IPv6 Gateway:

IPv6 DNS Server:

Assign the IP to the web server

Server2

Physical Config Services Desktop Programming Attributes

IP Configuration

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IP Address 192.168.1.6

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 192.168.1.3

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

Link Local Address FE80::290:CFF:FEE5:215B

IPv6 Gateway

IPv6 DNS Server

☐ Top

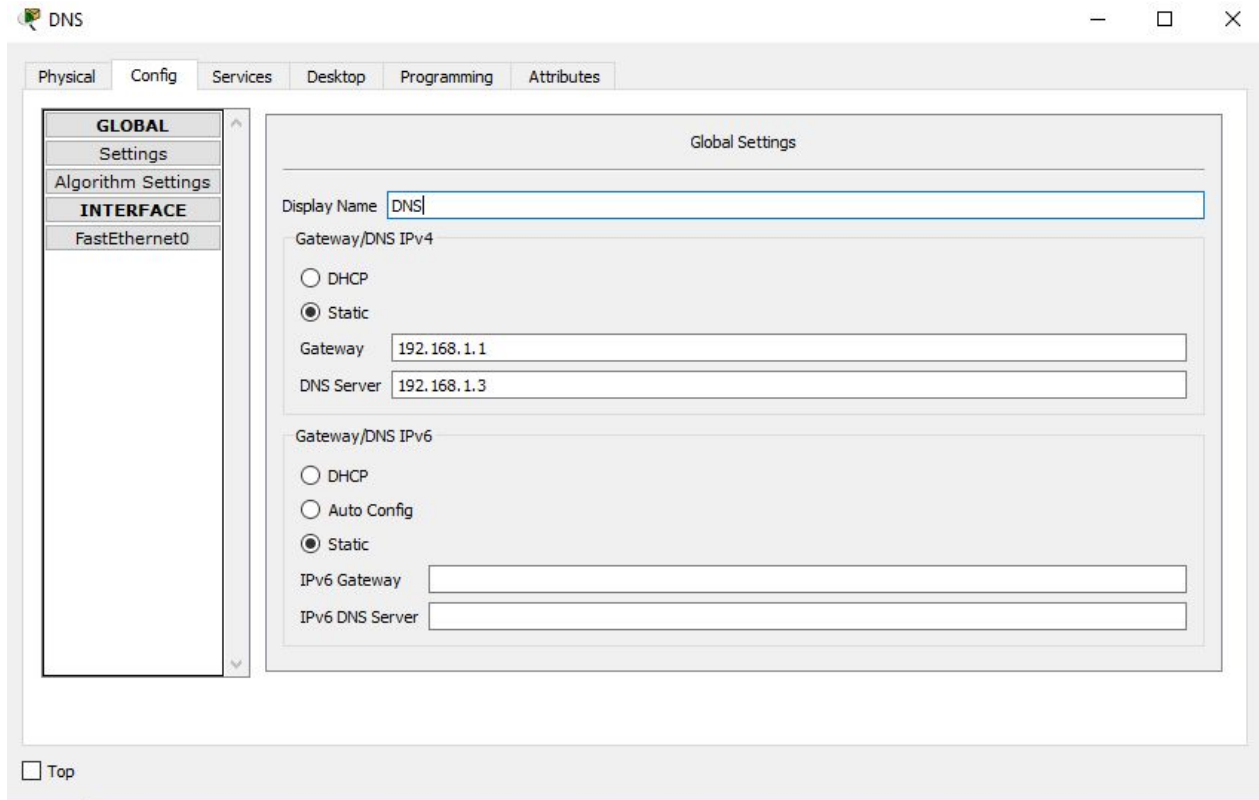
Go to first server, Rename it to DHCP

The screenshot shows a window titled "DHCP" with standard window controls (minimize, maximize, close) in the top right corner. The window has a tabbed interface with "Physical", "Config", "Services", "Desktop", "Programming", and "Attributes". The "Config" tab is active. On the left is a sidebar with a tree view containing "GLOBAL" (expanded), "Settings", "Algorithm Settings", "INTERFACE", and "FastEthernet0". The main area is titled "Global Settings" and contains the following configuration fields:

- Display Name:** A text field containing "DHCP".
- Gateway/DNS IPv4:** A section with two radio buttons: "DHCP" (unselected) and "Static" (selected). Below the radio buttons are two text fields: "Gateway" with the value "192.168.1.1" and "DNS Server" with the value "192.168.1.3".
- Gateway/DNS IPv6:** A section with three radio buttons: "DHCP" (unselected), "Auto Config" (unselected), and "Static" (selected). Below the radio buttons are two text fields: "IPv6 Gateway" and "IPv6 DNS Server", both of which are currently empty.

At the bottom left of the window, there is a "Top" button.

Go to first server, Rename it to DNS



The screenshot shows a window titled "DNS" with a standard Windows-style title bar (minimize, maximize, close buttons). The window has a tabbed interface with tabs labeled "Physical", "Config", "Services", "Desktop", "Programming", and "Attributes". The "Config" tab is selected. On the left side of the "Config" tab, there is a tree view with the following structure:

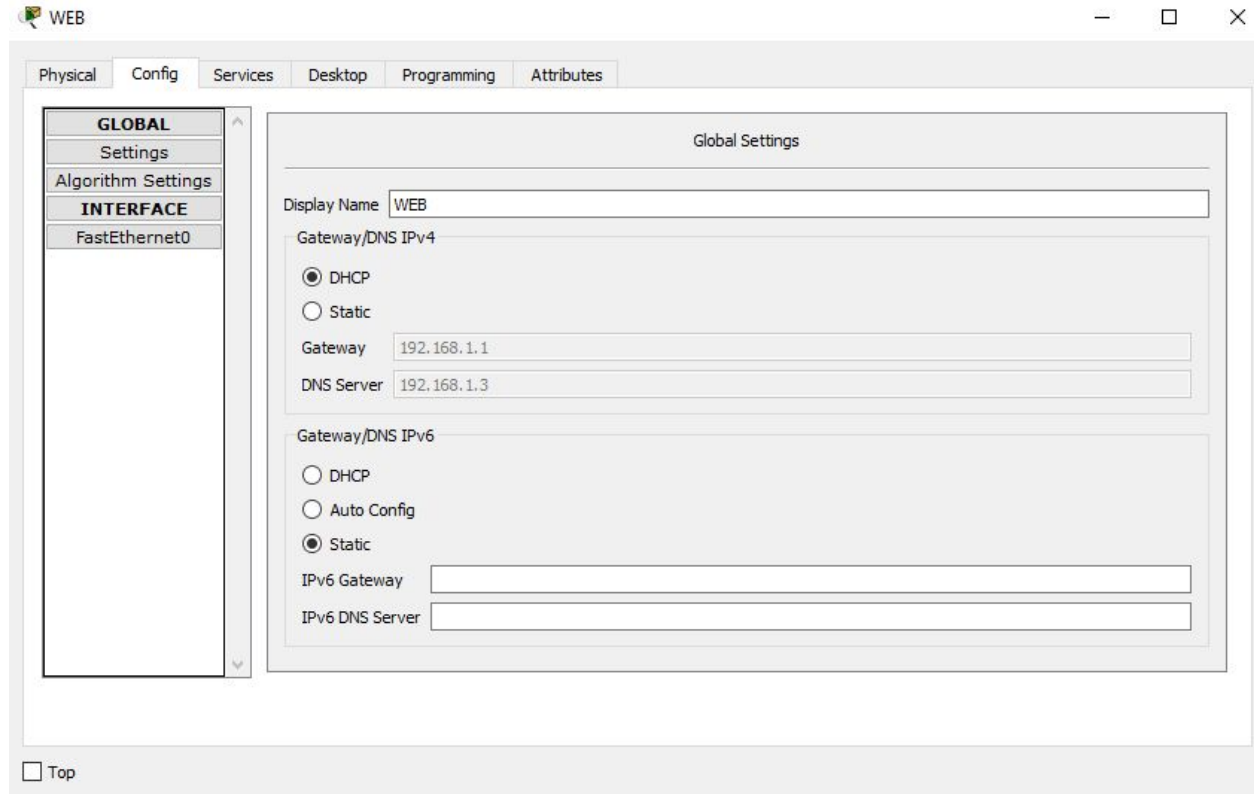
- GLOBAL**
 - Settings
 - Algorithm Settings
- INTERFACE**
 - FastEthernet0

The main area of the window displays the "Global Settings" for the DNS service. It contains the following fields and options:

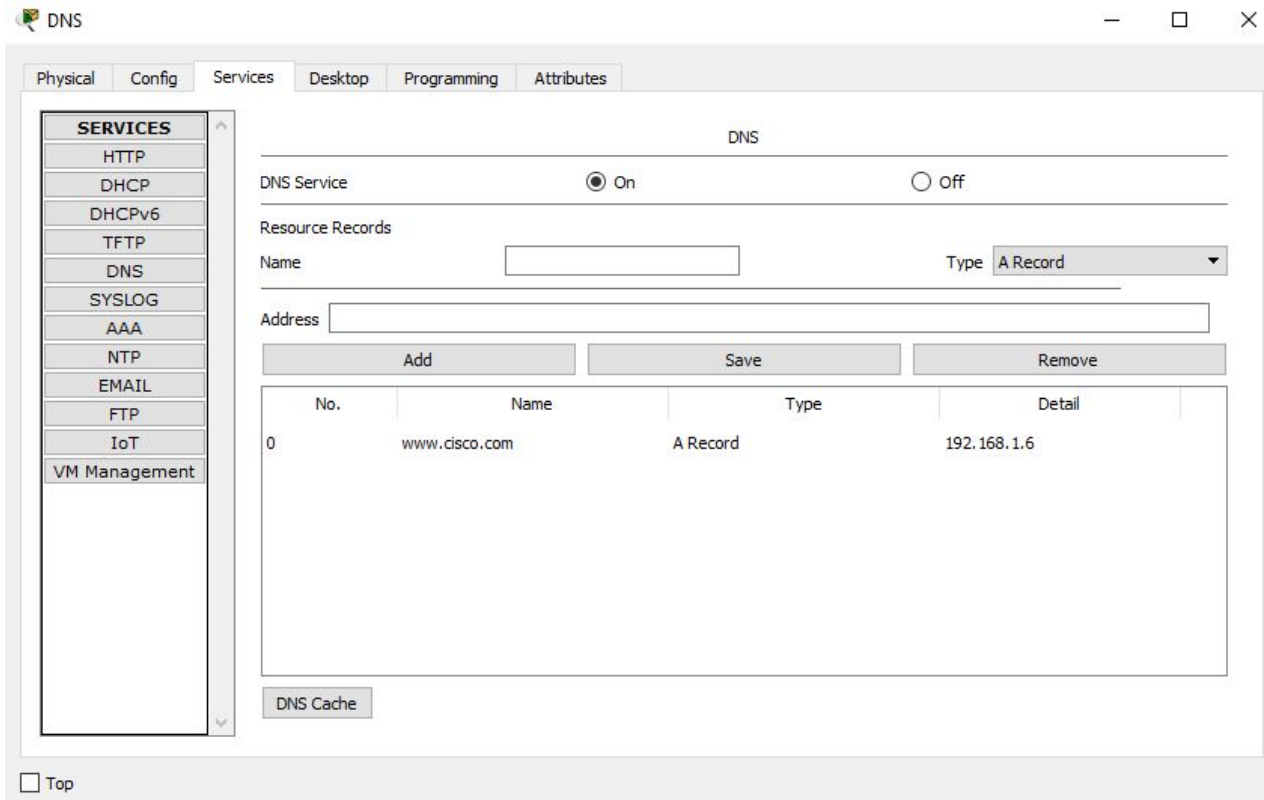
- Display Name:** A text field containing the value "DNS".
- Gateway/DNS IPv4:**
 - Radio buttons for ☐ DHCP and ☒ Static.
 - Gateway:** A text field containing the value "192.168.1.1".
 - DNS Server:** A text field containing the value "192.168.1.3".
- Gateway/DNS IPv6:**
 - Radio buttons for ☐ DHCP, ☐ Auto Config, and ☒ Static.
 - IPv6 Gateway:** An empty text field.
 - IPv6 DNS Server:** An empty text field.

At the bottom left of the window, there is a "Top" button with a small square icon next to it.

Go to third server, Rename it to WEB Server



Go to DNS Server > Services > Fill up the Name and Address > Click Add



The screenshot shows a web-based configuration interface for a DNS server. The 'Services' tab is active, displaying a list of services on the left and configuration options on the right. The 'DNS Service' is turned on. Under 'Resource Records', there is a table with one entry: 'www.cisco.com' with an 'A Record' type and IP address '192.168.1.6'. The interface includes buttons for 'Add', 'Save', and 'Remove' records, and a 'DNS Cache' button at the bottom.

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management

DNS

DNS Service ☒ On ☐ off

Resource Records

Name Type **A Record**

Address

Add **Save** **Remove**

No.	Name	Type	Detail
0	www.cisco.com	A Record	192.168.1.6

DNS Cache

☐ Top

Go to Web Server > Services > Select HTTP > Click Edit

The screenshot shows a web configuration interface titled "WEB". It has several tabs: Physical, Config, Services, Desktop, Programming, and Attributes. The "Services" tab is selected. On the left, there is a "SERVICES" sidebar with a list of services: HTTP, DHCP, DHCPv6, TFTP, DNS, SYSLOG, AAA, NTP, EMAIL, FTP, IoT, and VM Management. The "HTTP" service is selected. In the main area, there are two sections: "HTTP" and "HTTPS". Each section has a radio button for "On" (selected) and "Off". Below these is a "File Manager" table with columns for File Name, Edit, and Delete. The table lists five files: copyrights.html, cscoptlogo177x111.jpg, helloworld.html, image.html, and index.html. At the bottom right, there are "New File" and "Import" buttons. A "Top" button is at the bottom left.

WEB

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management

HTTP

HTTP ☒ On ☐ Off

HTTPS ☒ On ☐ Off

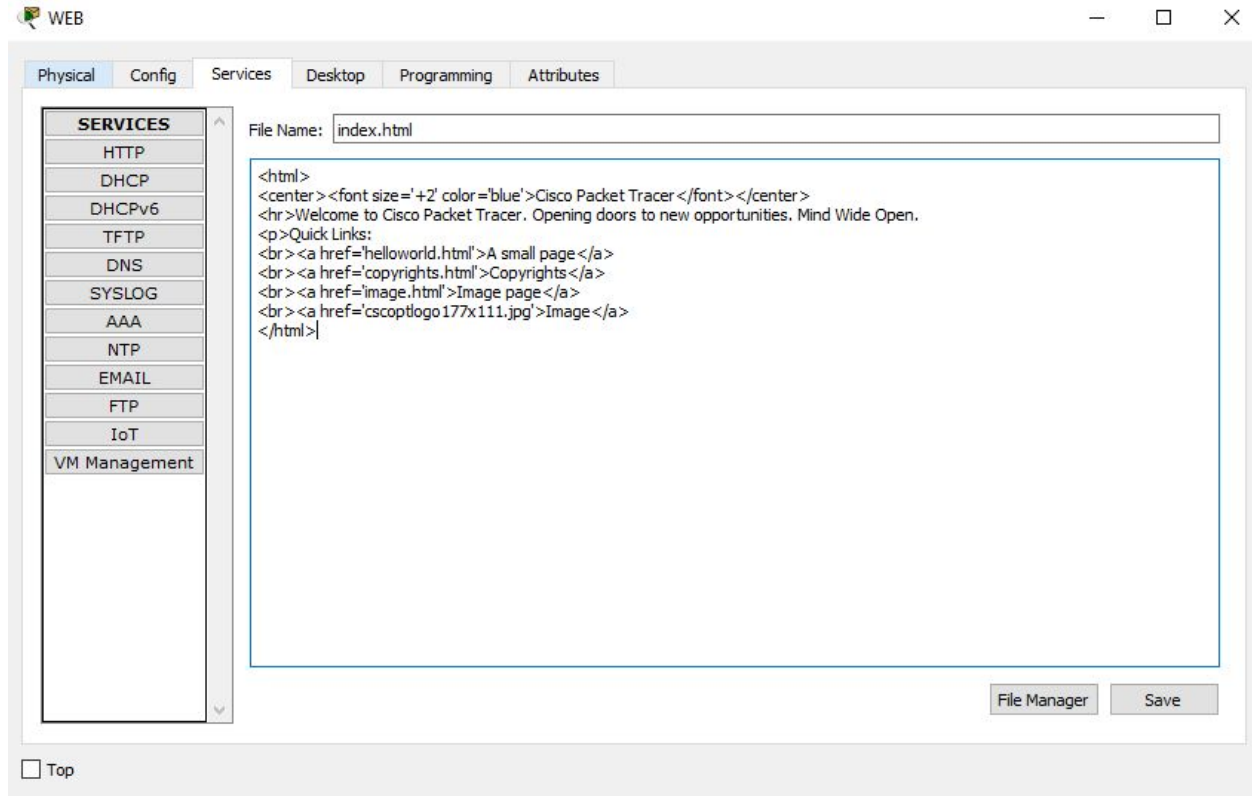
File Manager

	File Name	Edit	Delete
1	copyrights.html	(edit)	(delete)
2	cscoptlogo177x111.jpg		(delete)
3	helloworld.html	(edit)	(delete)
4	image.html	(edit)	(delete)
5	index.html	(edit)	(delete)

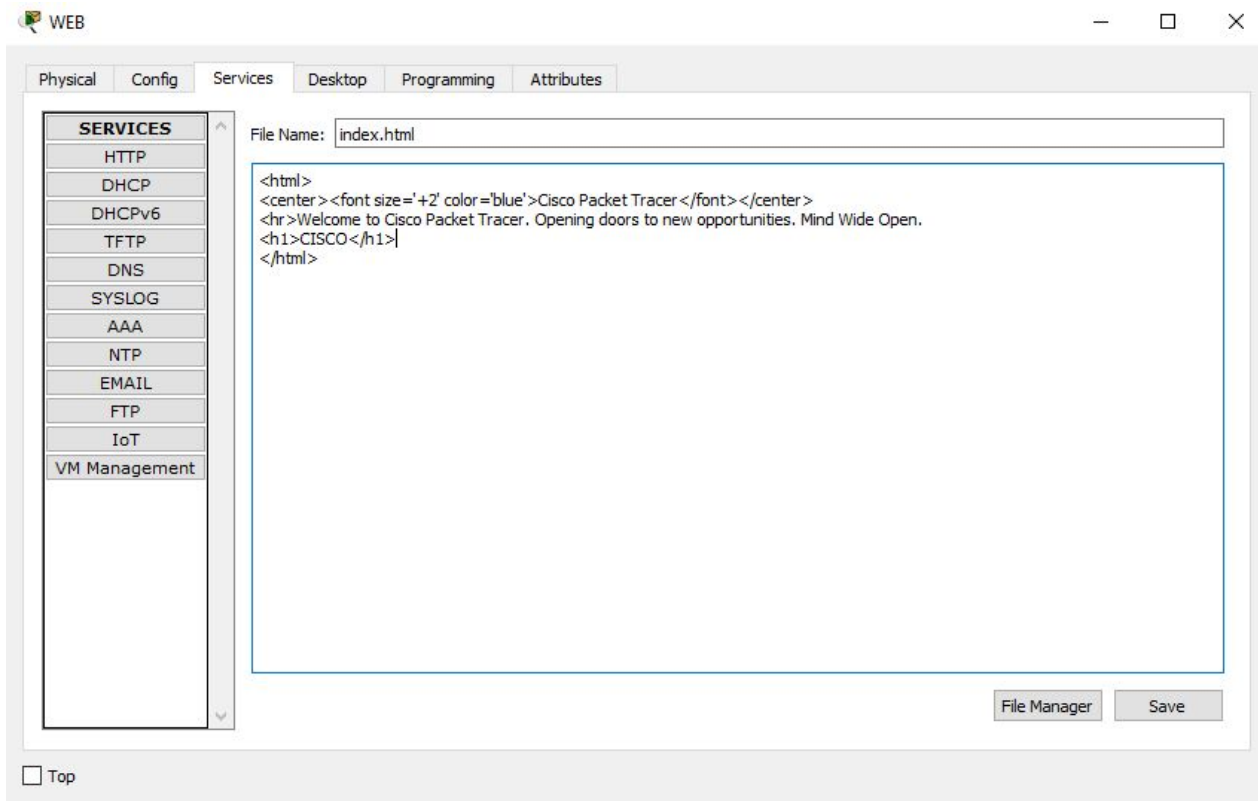
New File Import

☐ Top

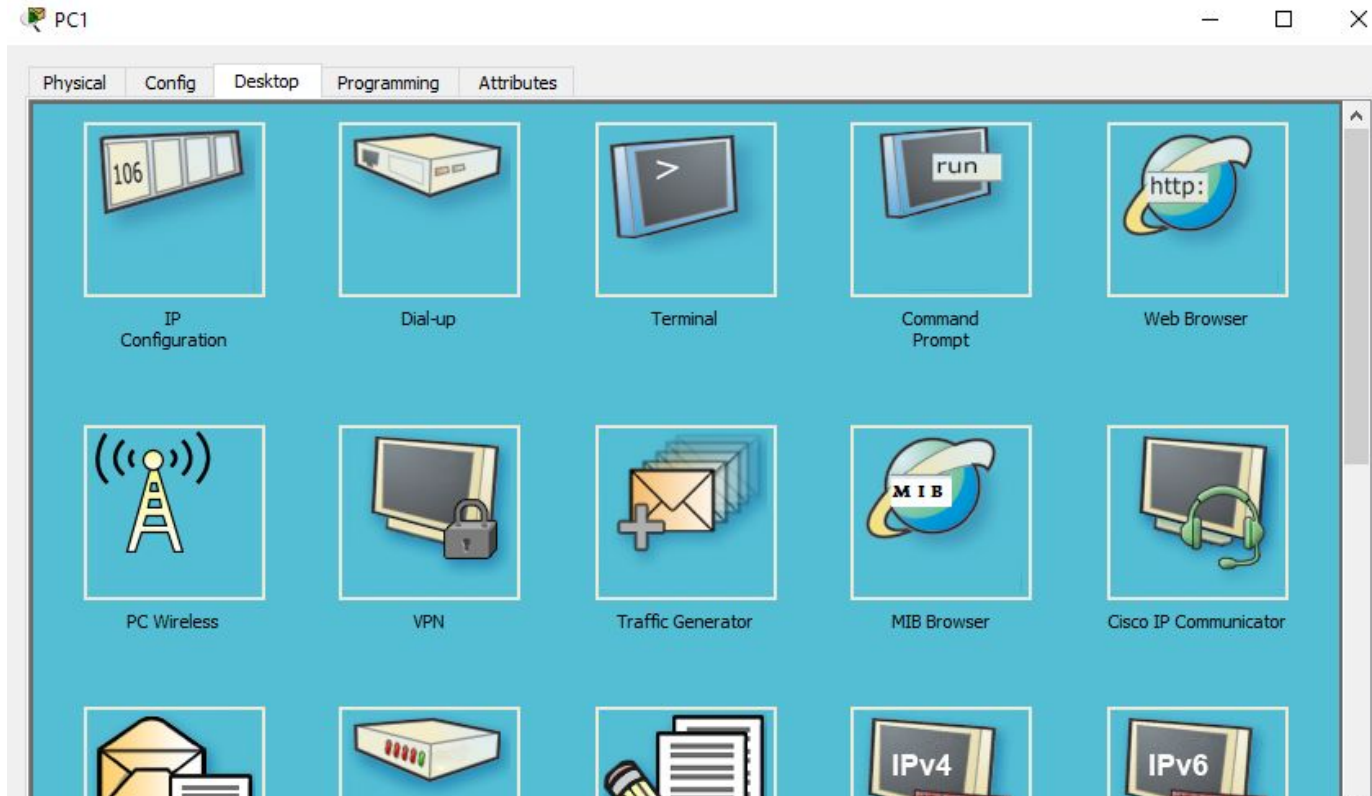
After Clicking You will get this page Now edit the web page



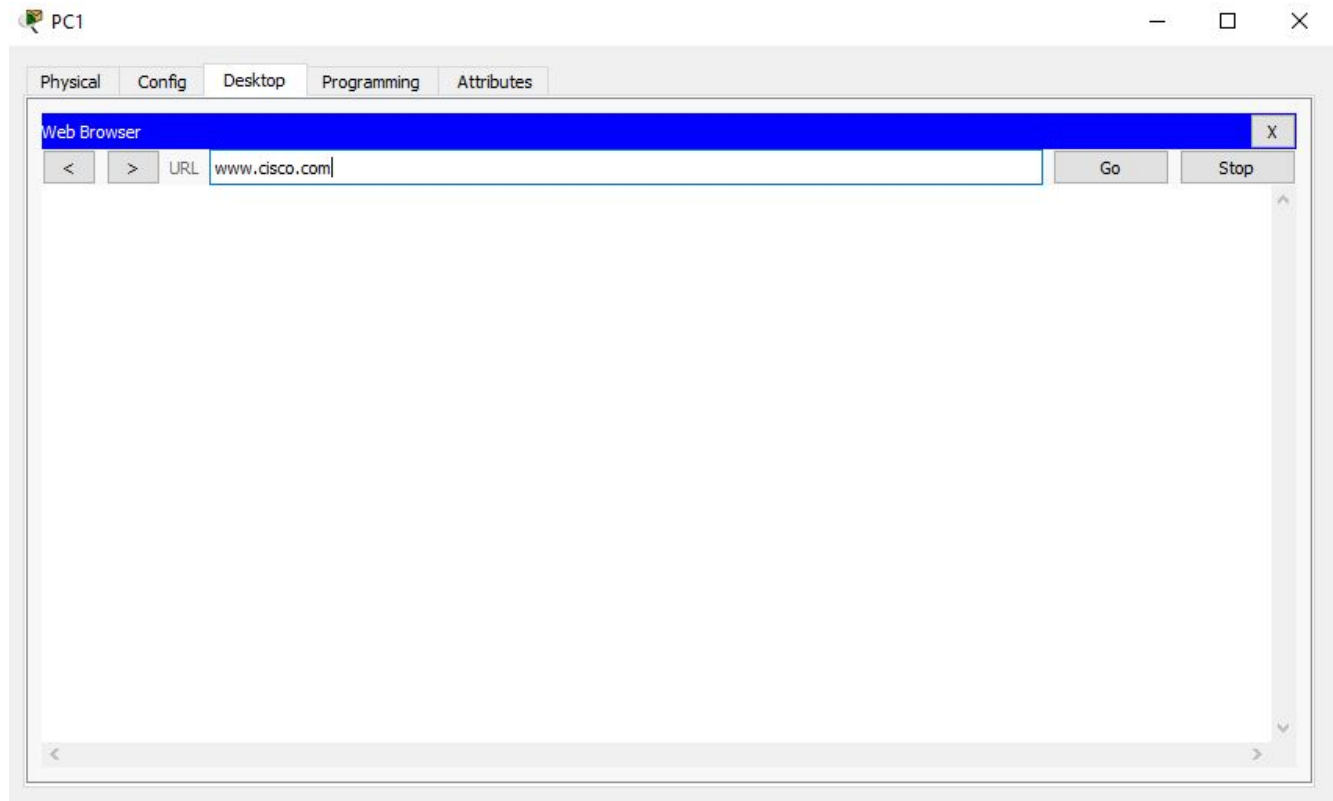
After editing the web page, click save



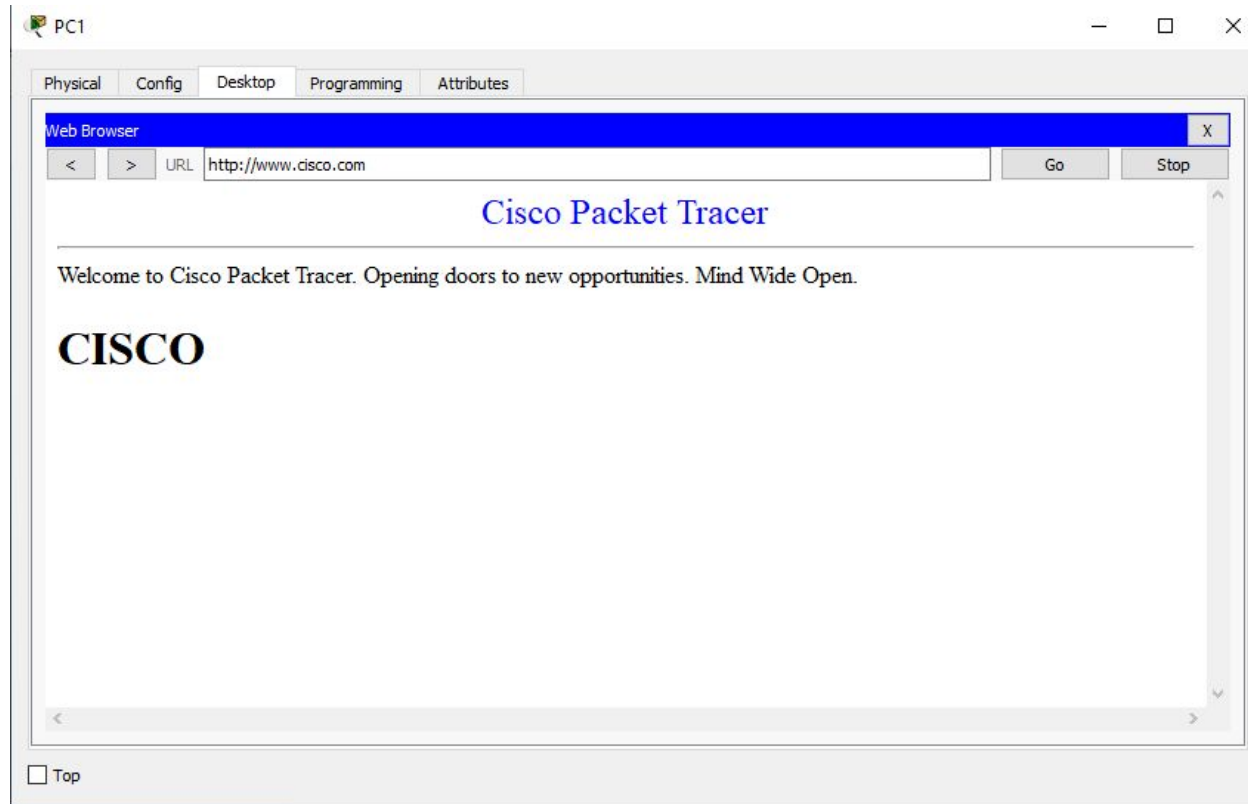
From PC1 Click the web browser



Type the IP or the corresponding Domain Name, Clock Go



Content of the site should be visible



Content of the site should be visible even if the ip of the web server is given

